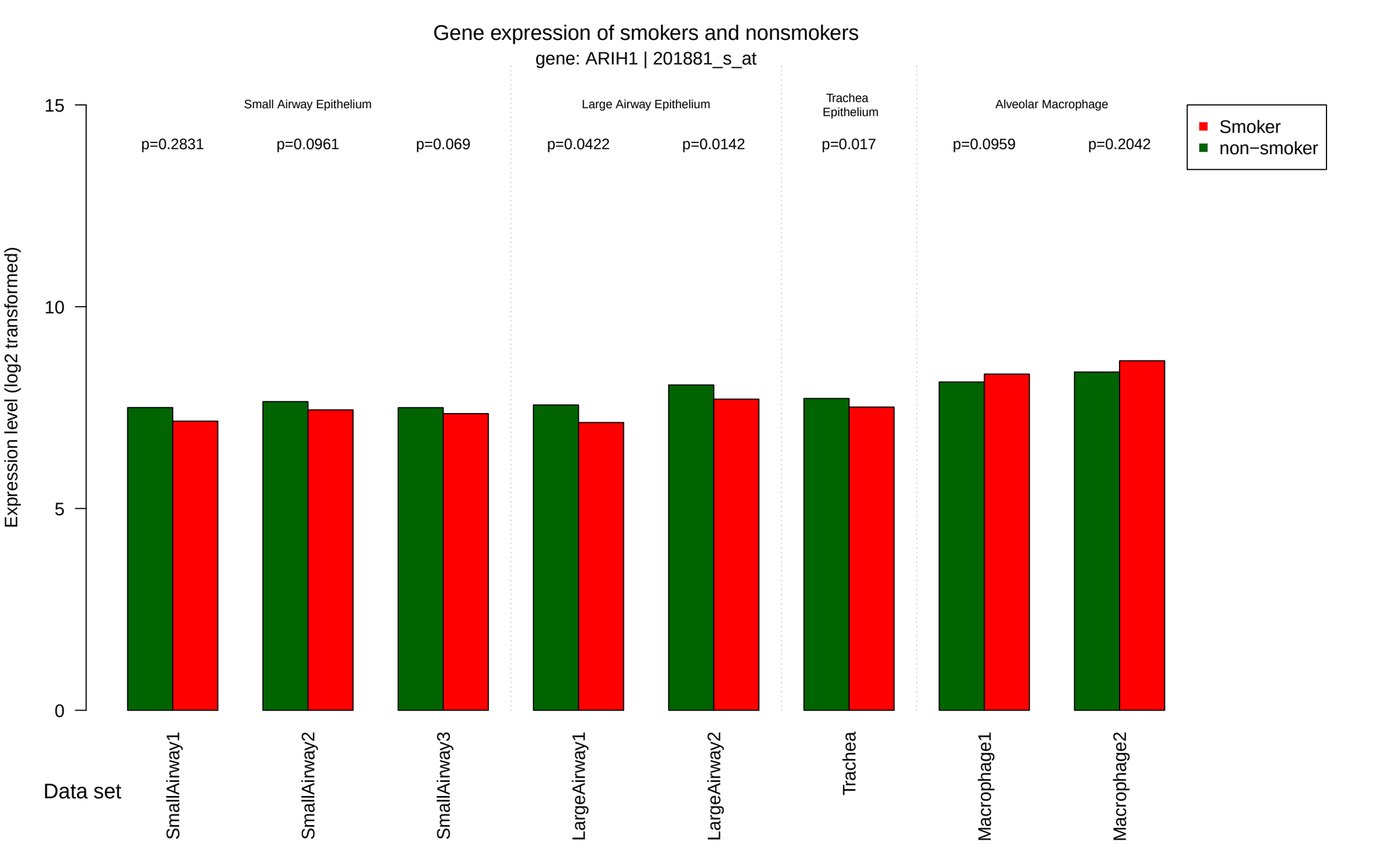
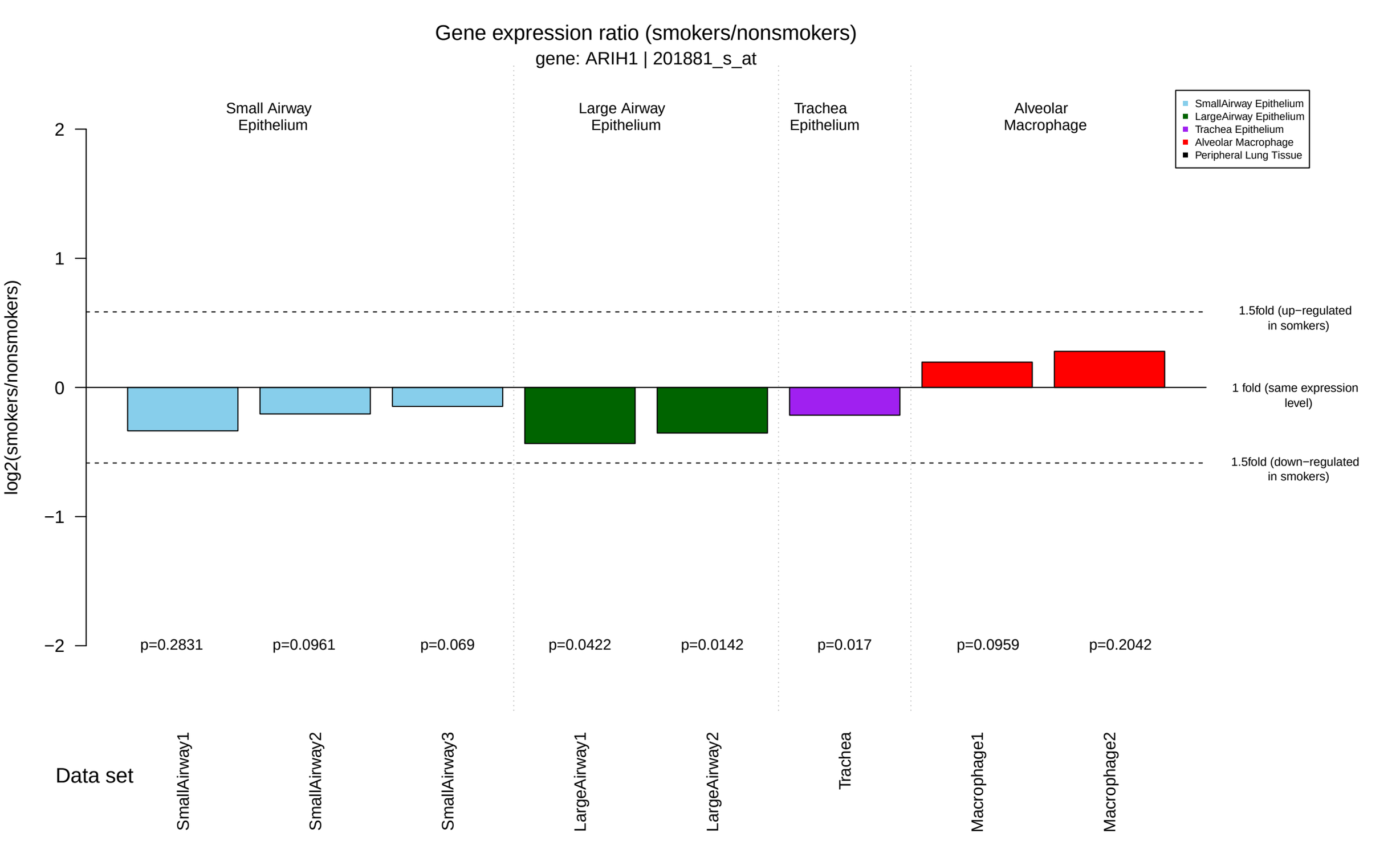
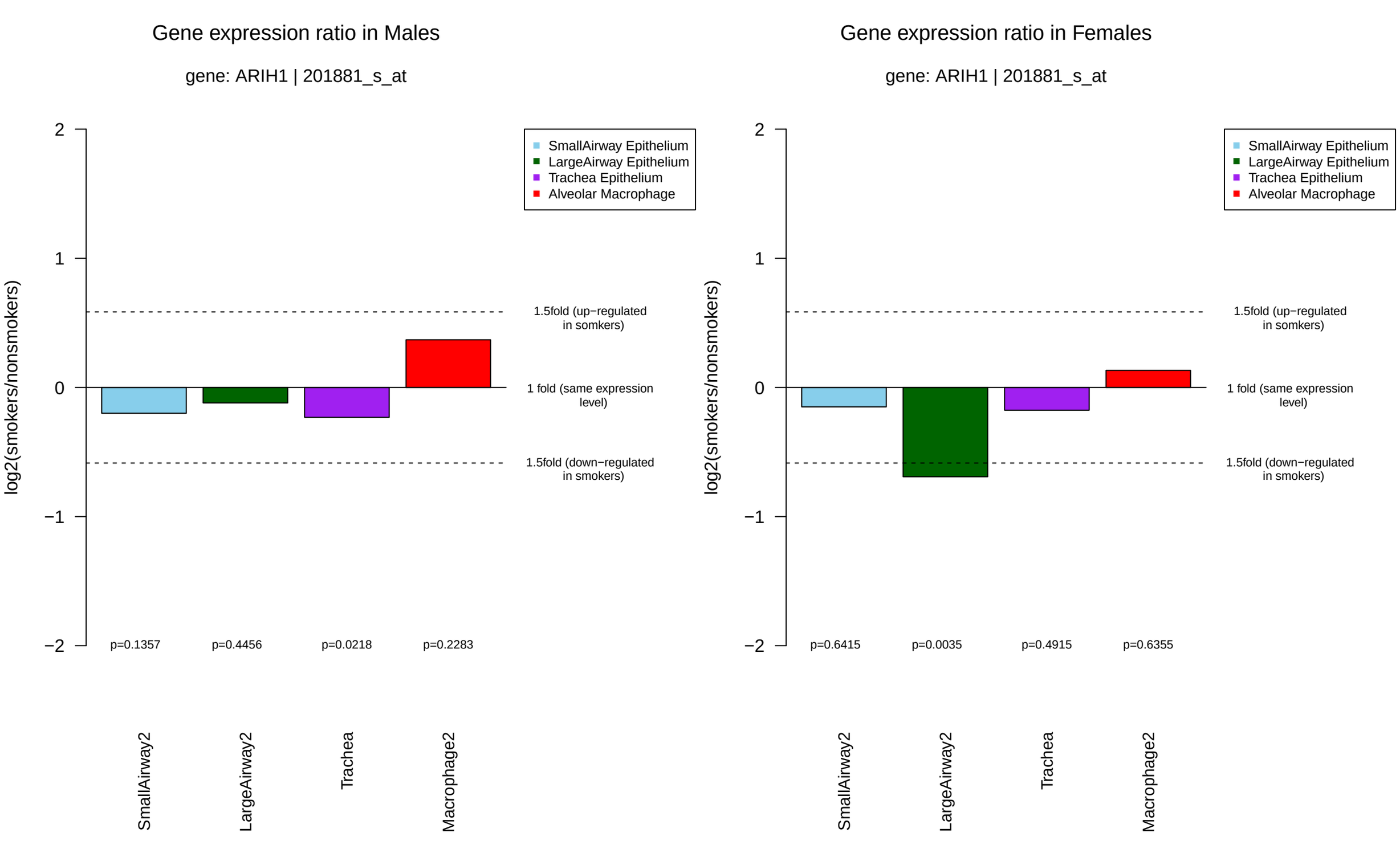
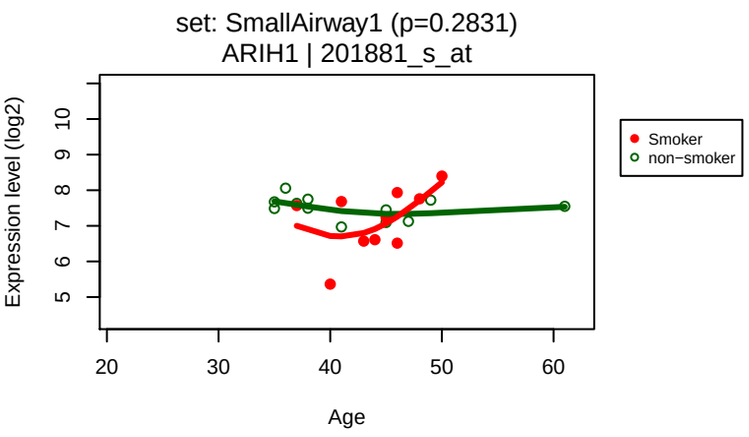
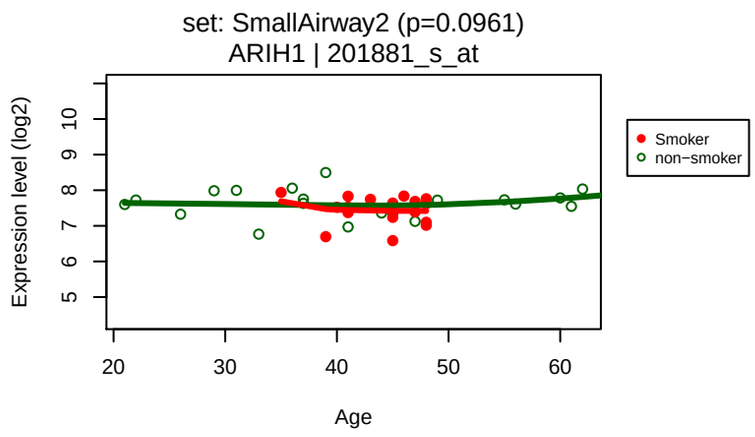
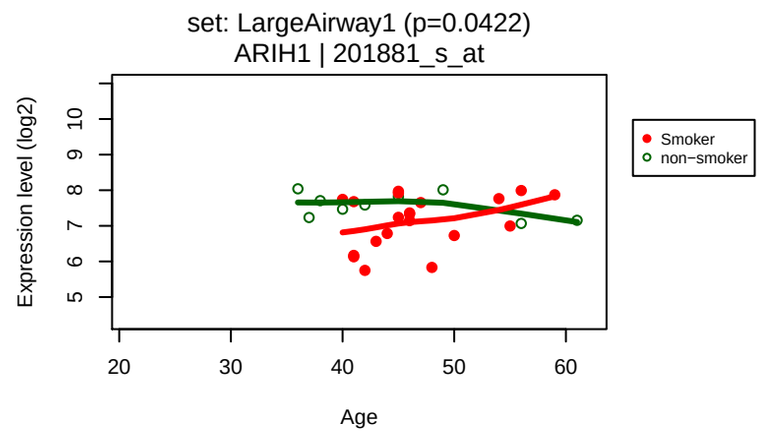

Gene expression of smokers and nonsmokers
gene: ARIH1 | 201881_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2831
p=0.0961
p=0.069
p=0.0422
p=0.0142
p=0.017
p=0.0959
p=0.2042
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

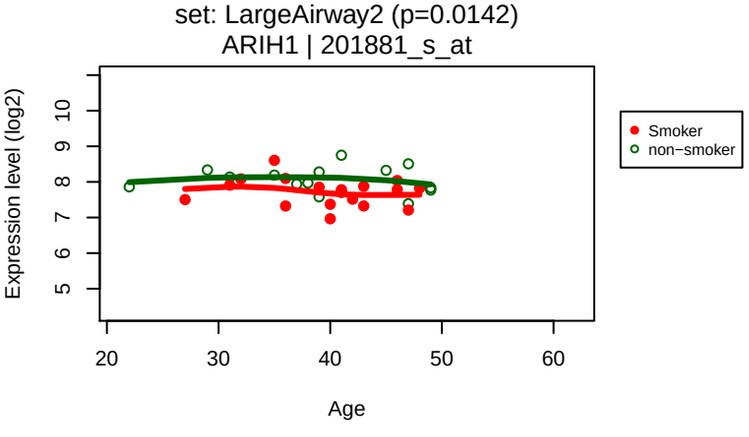

Gene expression ratio (smokers/nonsmokers)
gene: ARIH1 | 201881_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2831
p=0.0961
p=0.069
p=0.0422
p=0.0142
p=0.017
p=0.0959
p=0.2042
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

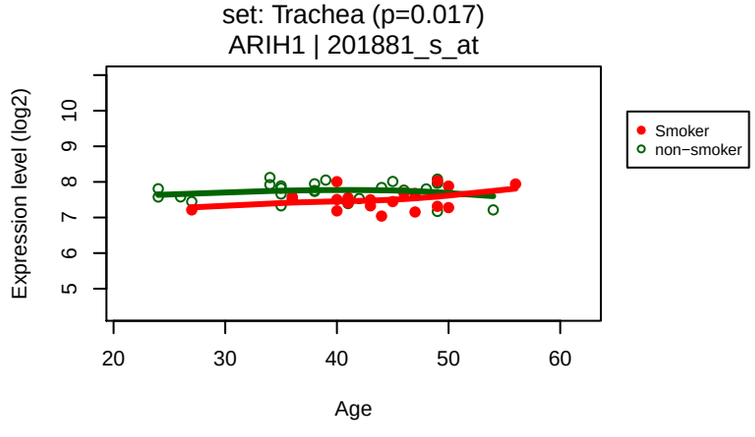

Gene expression ratio in Males
gene: ARIH1 | 201881_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1357
p=0.4456
p=0.0218
p=0.2283
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARIH1 | 201881_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6415
p=0.0035
p=0.4915
p=0.6355
SmallAirway2
LargeAirway2
Trachea
Macrophage2

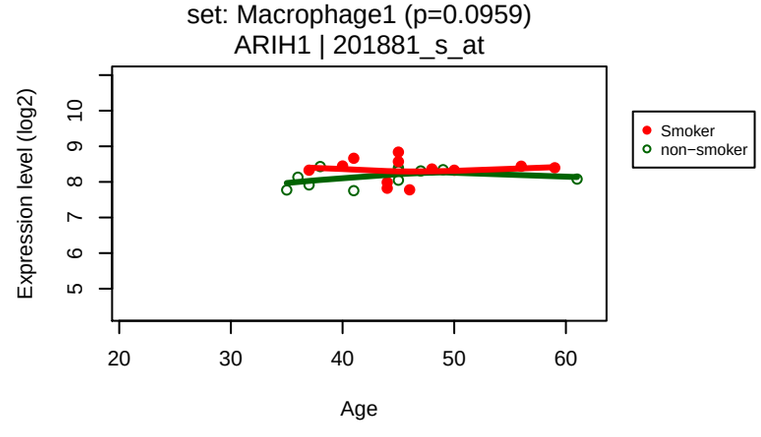

set: SmallAirway1 (p=0.2831)
ARIH1 | 201881_s_at
Expression level (log2)
Age
Smoker
non−smoker

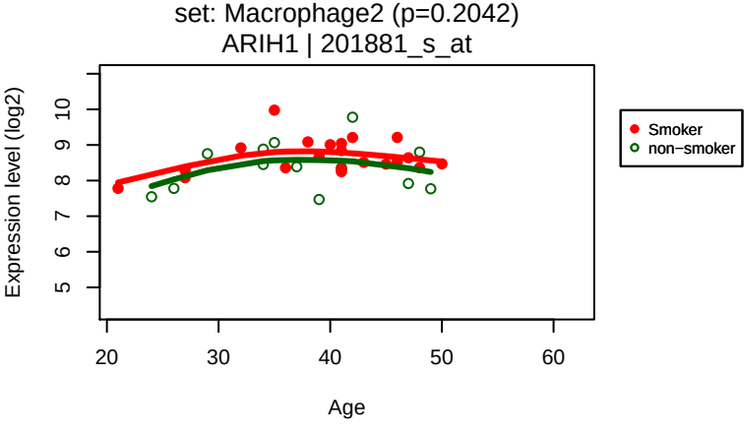

set: SmallAirway2 (p=0.0961)
ARIH1 | 201881_s_at
Expression level (log2)
Age
Smoker
non−smoker

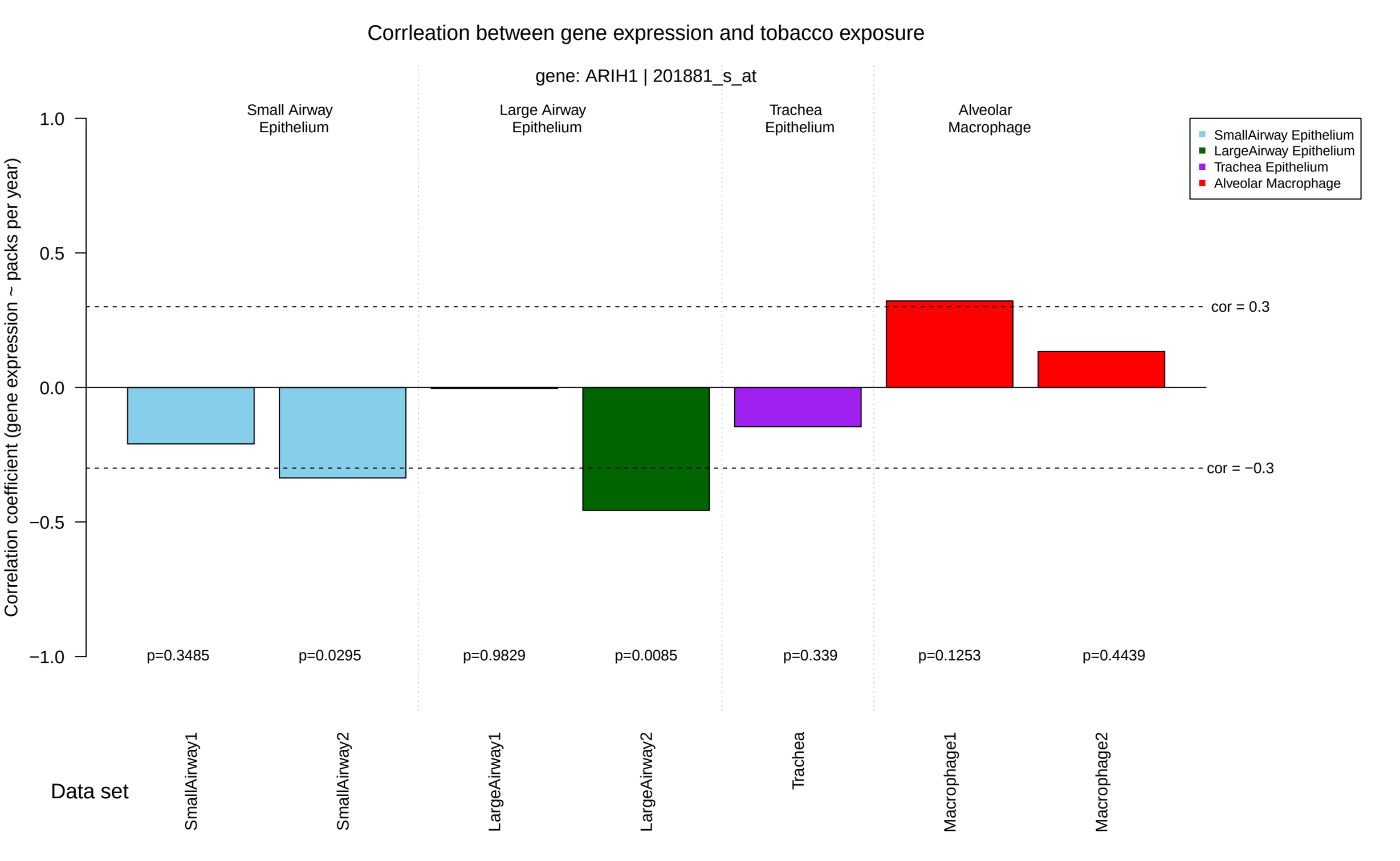

set: LargeAirway1 (p=0.0422)
ARIH1 | 201881_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0142)
ARIH1 | 201881_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.017)
ARIH1 | 201881_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0959)
ARIH1 | 201881_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2042)
ARIH1 | 201881_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARIH1 | 201881_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3485
p=0.0295
p=0.9829
p=0.0085
p=0.339
p=0.1253
p=0.4439
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2